Teaching a student who forgets: one-page summary

Thesis work on adaptive tutoring (cursim simulator). Results from 2026-10-18.

What we are studying

Students forget. A topic learned today is partly gone after a break. A tutor picks which question to ask next, and we ask: **if we know how long the break will be, how should the tutor change what it does?** The break length is not something the tutor controls; it is the situation. We look at two things the tutor can change.

- **Revision budget:** what share of the questions go to topics already learned.
- **Study depth:** how many right answers in a row the tutor wants before it moves on to the next topic.

How we tested it

A simulated student (40 made-up topics, with prerequisites) practises for 20 days, 25 questions a day. Then there is a break of 0 to 28 days. We count the topics the student can still use after the break (right more than half the time; the maximum is 40). We tried three memory speeds (half-life 7, 14 and 28 days). The best setting for each break was picked on 50 students and scored on 100 new ones.

The tutors

The base is **CurriculumTutor**, now coded exactly as in the paper (it picks topics with a bandit rule and decides a topic is learned with a change-detection test, needing 7 right answers in a row). Every other tutor is that same tutor with **one or two settings changed**: revision budget 0 is CurriculumTutor itself, and depth 7 in a row is the paper's depth. Nothing else differs. We found that our earlier "CurriculumTutor" was a stand-in, not the paper's algorithm, so the old comparisons are retired and adaptive review is dropped.

What we found (students with a 14-day memory half-life)

- **The published tutor is far from the best.** After a 3-day break it keeps 15.4 topics; the best settings keep 33.6. On average over breaks it is 17 topics short.
- **Revision budget:** the longer the break, the more revision is best (0.1 with no break, 0.3 at 3 days, 0.4 at 7 days, 0.5 at 10 days).
- **Study depth on its own does nothing.** The paper's depth is already about right when there is no revision.
- **Together they work:** with revision, "learn it quickly (about 3 in a row), then revise" beats "learn it deeply". Depth only needs to grow when the break is long (8 in a row at 10 days).
- **No single setting is best for every break.** The best fixed revision budget still loses 8.1 topics on average; the best fixed pair loses 3.4 on average and 10.3 at worst, against 17.4 for the published tutor.
- Same pattern for 7-day students (revise 0.4 to 0.5 at once) and 28-day students (no revision until about 7 days).

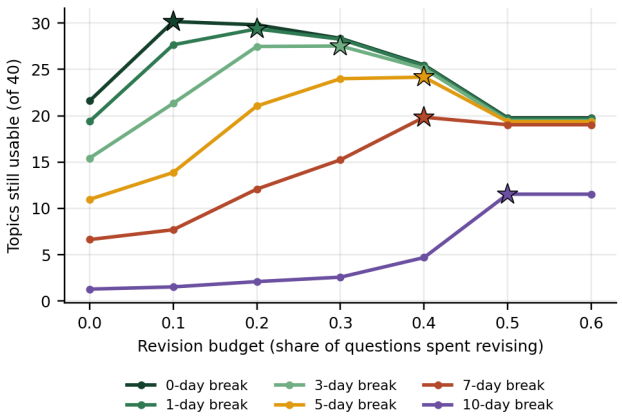


Chart: topics still usable after the break, against the share of questions spent revising (stars = best for that break; depth as published).

Topics usable after the break (14-day half-life)

Break	Published tutor	Best revision budget alone	Best revision + depth
0 d	21.6	30.2 (budget 0.1)	39.1 (budget 0.4, 3 in a row)
1 d	19.4	29.4 (budget 0.2)	38.3 (budget 0.4, 3 in a row)
3 d	15.4	27.5 (budget 0.3)	33.6 (budget 0.4, 4 in a row)
5 d	11.0	24.2 (budget 0.4)	29.9 (budget 0.5, 3 in a row)
7 d	6.6	19.8 (budget 0.4)	23.9 (budget 0.5, 4 in a row)
10 d	1.3	11.5 (budget 0.5)	11.7 (budget 0.5, 8 in a row)

Limits and next steps

Simulated students only, on a made-up curriculum; no real-student data yet. The best setting for each break is slightly flattered because it was chosen on a small group (then scored on fresh students). Which topic to revise (least recently asked) and the tutor's guess and slip values are our choices. **Next:** a tutor that sets its revision budget and depth without being told the break length, then a test on real learner logs. Code and numbers: results/knob_sweep and experiment_logs.md.